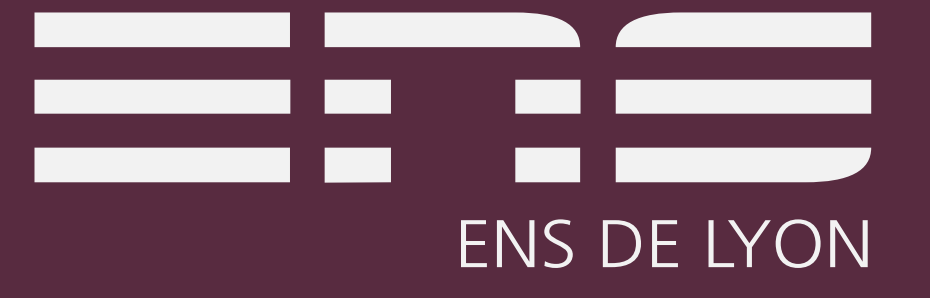


Equivariant imaging: self-supervised learning for phase retrieval

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Summary / TLDR

Goal: Learn to reconstruct signals from phaseless measurements alone.

Challenges:

1. When is learning possible?
2. How can we train a network with degraded measurements only?

Self-supervised learning for phase retrieval problem

Problem setup:

$$\mathbf{y} = h(\mathbf{x}) := |\mathbf{A}\mathbf{x}|^2$$

where:

- $\mathbf{x}$ belongs to a set $\mathcal{X} \subset \mathbb{C}^n$.
- $\mathbf{A} \in \mathbb{C}^{m \times n}$ is a linear operator (e.g. Fourier transform, random matrix, etc.).
- $|\cdot|$ modulus element-wise **non-linear** operator.

Self-supervised learning:

- **Learn a reconstruction function** $f: \mathbf{y} \mapsto \mathbf{x}$, from a dataset of measurement data only $\{\mathbf{y}_i\}_{i \in I}$.
- $\alpha = m/n$ is the **sampling rate**.

Transformation invariance prior:

- Set of images is **invariant** to 2D shifts.
- **Shift invariance** enables learning from phaseless data.

Why invariance is needed?

We cannot learn to reconstruct $\mathbf{x}$ from $\mathbf{y}$ since $f(\mathbf{y}) = \mathbf{A}^\dagger \text{diag}(e^{i\phi_1}, \dots, e^{i\phi_m}) \sqrt{\mathbf{y}}$ is a measurement consistent solution: $|\mathbf{A}f(\mathbf{y})|^2 = \mathbf{y}$, for any $(\phi_1, \dots, \phi_m)^\top \in \mathbb{R}^m$.

Metric: Cosine Similarity (CS)

$$\text{CS}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{|\mathbf{x}^* \hat{\mathbf{x}}|}{\|\mathbf{x}\|_2 \|\hat{\mathbf{x}}\|_2} \quad \begin{cases} \text{CS} \approx 1 \Leftrightarrow \hat{\mathbf{x}} \approx r e^{i\varphi} \mathbf{x}, & \text{with } r \in \mathbb{R} \text{ and } \varphi \in \mathbb{R}. \\ \text{CS} \approx 0 \Leftrightarrow \mathbf{x} \perp \hat{\mathbf{x}} \end{cases}$$

$$\hat{\mathbf{x}} \stackrel{\varphi}{\approx} \mathbf{x} \text{ if } \exists r \in \mathbb{R}, \varphi \in \mathbb{R}, \text{ such that } \hat{\mathbf{x}} = r e^{i\varphi} \mathbf{x}.$$

Exploiting shift invariance in the network f_θ

Shift invariance:

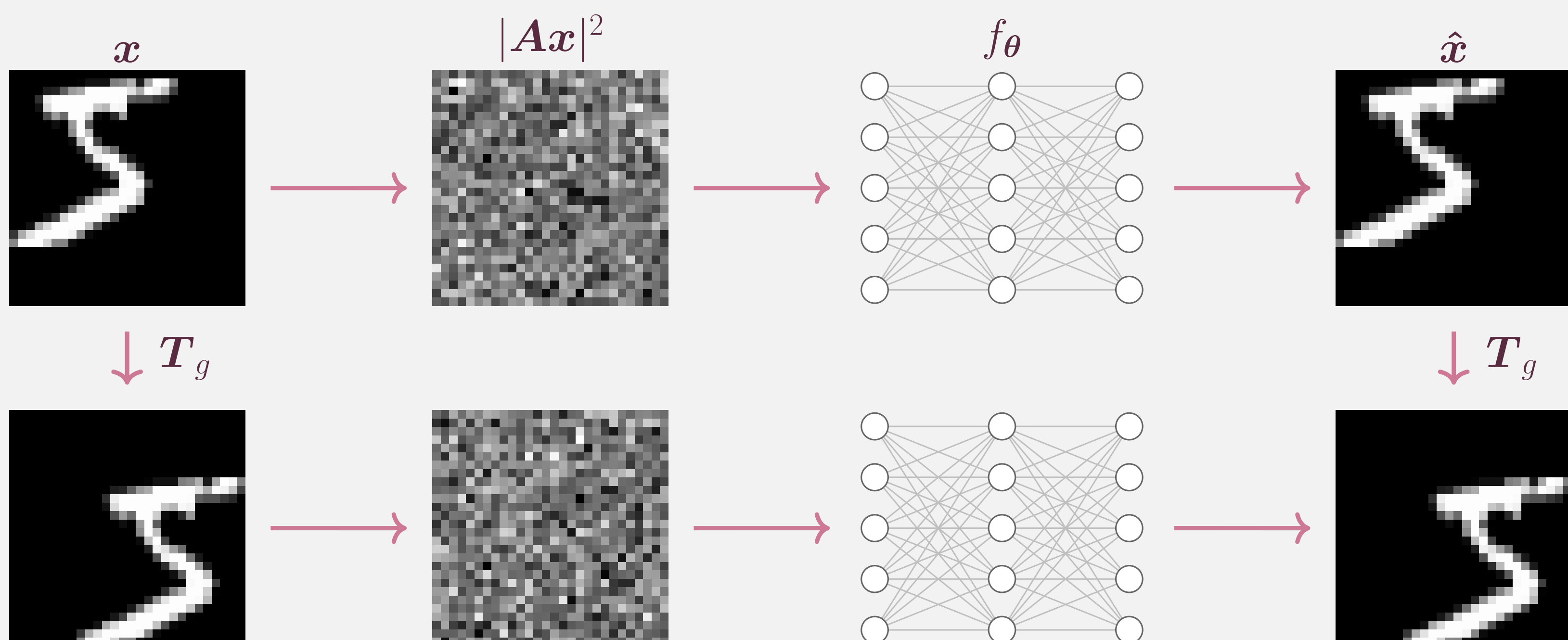
$$\forall \mathbf{x} \in \mathcal{X}, \forall g \in G, T_g \mathbf{x} \in \mathcal{X}.$$

Exploiting invariance: $\mathbf{x}, T_g \mathbf{x} \in \mathcal{X}$

$$f_\theta(h(T_g \mathbf{x})) \stackrel{\varphi}{\approx} T_g \mathbf{x} \quad \text{and} \quad f_\theta(h(\mathbf{x})) \stackrel{\varphi}{\approx} \mathbf{x}.$$

$\Rightarrow$ **Equivariance property:** $f \circ h$ equivariant for the group G :

$$\forall \mathbf{x} \in \mathcal{X}, g \in G: f_\theta(h(T_g \mathbf{x})) \stackrel{\varphi}{\approx} T_g f_\theta(h(\mathbf{x})).$$



Proposed method

Measurement consistency loss:

$$\mathcal{L}_A(\theta) = \sum_{i \in I} \left\| \sqrt{\mathbf{y}_i} - \sqrt{h(f_\theta(\mathbf{y}_i))} \right\|^2$$

$$\mathcal{L}_I(\theta) = \sum_{i \in I} \left\| \mathbf{y}_i - h(f_\theta(\mathbf{y}_i)) \right\|^2$$

Equivariance promoting loss:

$$\mathcal{L}_{EI}(\theta) = - \sum_{\substack{i \in I \\ g \in G}} \text{CS}(T_g f_\theta(\mathbf{y}_i), f_\theta(h(T_g f_\theta(\mathbf{y}_i))))$$

$$\text{Training loss: } \mathcal{L}(\theta) = \mathcal{L}_{AI}(\theta) + \lambda \mathcal{L}_{EI}(\theta).$$

Experiments

► **“Phase MNIST” dataset:**

$\mathbf{x} = e^{-i\mathbf{x}_0}$ where $\mathbf{x}_0$ is an MNIST image.

► $\mathbf{A} \in \mathbb{C}^{m \times n}$ **random** matrix with:

$$a_{k,l} \sim \mathcal{N}(0, 1/2m) + i\mathcal{N}(0, 1/2m).$$

► Measurements: $\mathbf{y} = |\mathbf{A}\mathbf{x}|^2$.

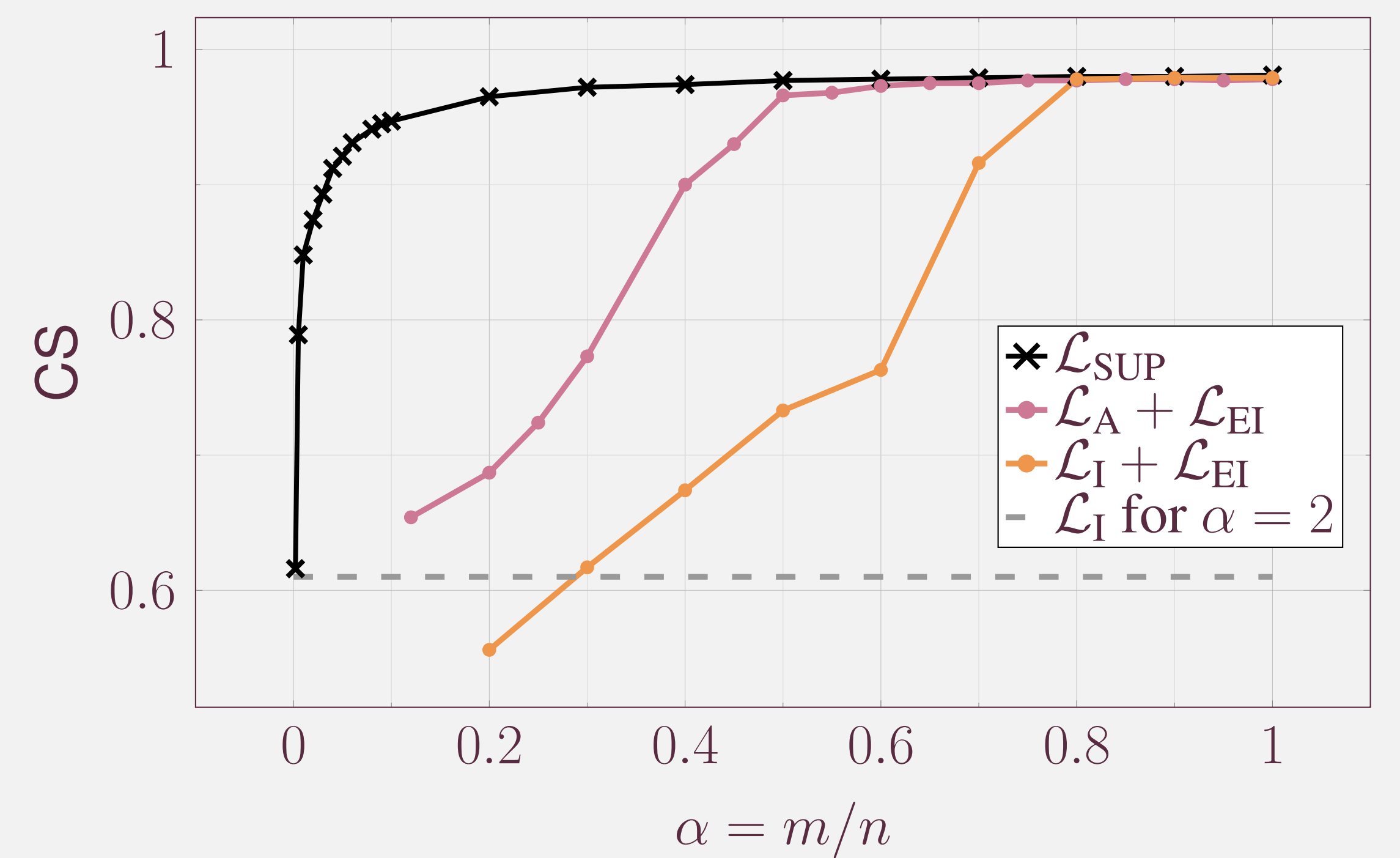
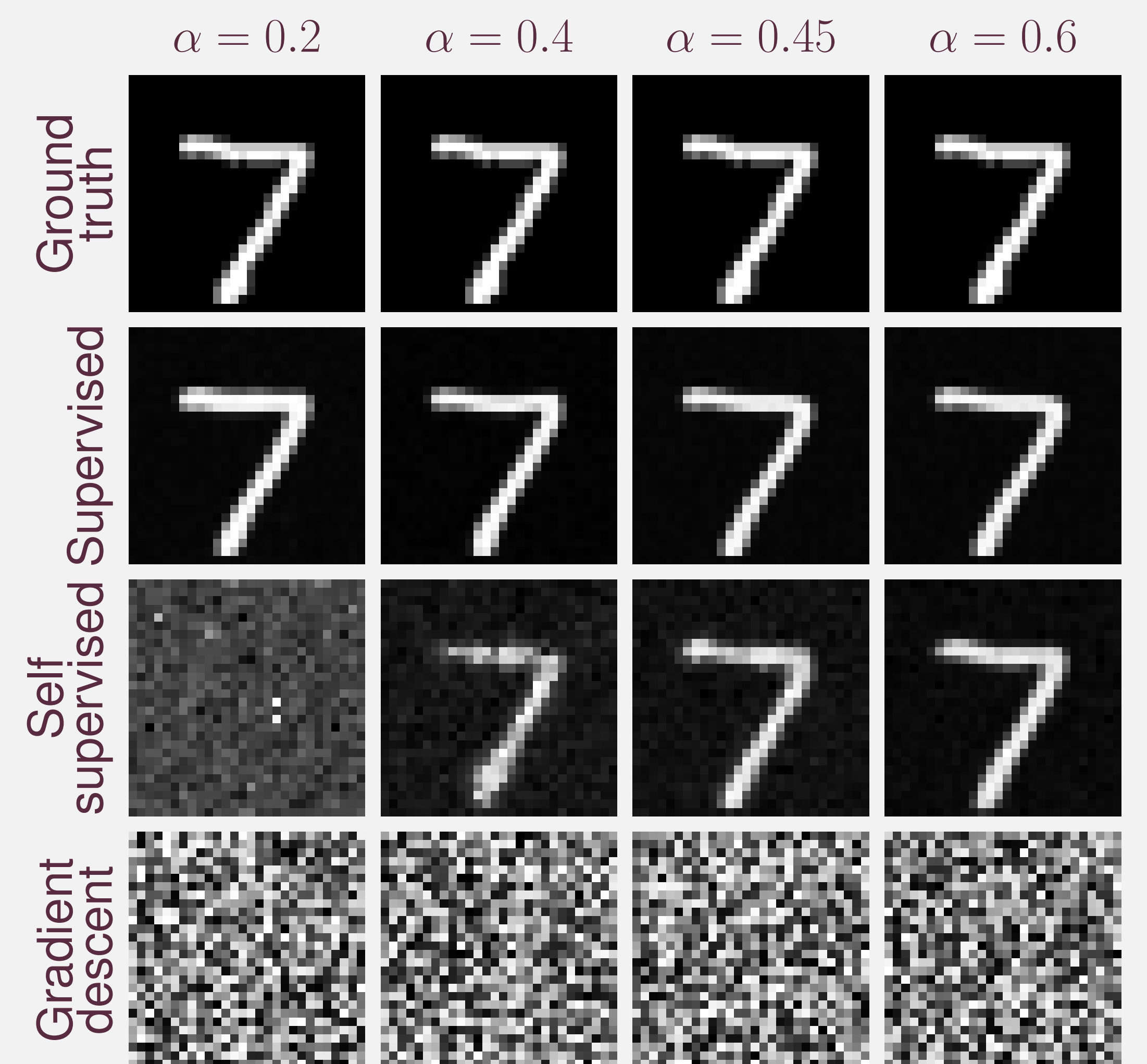


Figure 1: Cosine similarity as a function of sampling rate α for supervised and self-supervised methods.



Implementation details

- $f_\theta(\mathbf{y}) = g_\theta(\mathbf{A}^\top \mathbf{y})$ where g_θ is a U-Net architecture.
- **Loss parameter** $\lambda = 1$.
- **Two shifts** per image ($\mathcal{L}_{EI}$ loss).
- Optimization: **Learning rate:** 5×10^{-5} , **epochs:** 15, **Batch size:** 5.

Future works

- **Improve** the performance for **low** sampling rate.
- **Adapt** the method to other linear forward operators (e.g. **Fourier** transform).
- Provide **theoretical conditions** for learning depending on the **signal set** $\mathcal{X}$ and **sampling rate** α .

References

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